

TITLE PAGE

- **Problem Statement ID** – Question 1.
- **Problem Statement Title**- Eco Loop Building Agents
- **Theme**- Smart Automation / Sustainability / AI
- **PS Category**- Software
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- **Student ID**- 23BCE1803

IDEA TITLE

Proposed Solution:

Problem

Commercial buildings consume excessive electricity because HVAC systems operate using fixed schedules and rule-based controllers that cannot adapt to changing environmental conditions.

Our Solution

Eco-Loop is a **Physical AI system** that continuously monitors building conditions and automatically optimizes HVAC cooling/heating setpoints using an AI agent. Instead of manually controlling temperatures, the system:

- Reads live building sensor data
- Understands occupancy and weather
- Uses an LLM to make intelligent HVAC decisions
- Applies changes directly into a running EnergyPlus simulation
- Continuously learns every timestep

Key Features

- Live closed-loop HVAC control
- Real-time EnergyPlus simulation
- AI Tool Calling
- Dynamic cooling & heating optimization
- Occupancy-aware control
- Outdoor weather awareness
- Energy monitoring dashboard
- Carbon-aware optimization support

Innovation

- Uses **LLMs for real-time HVAC control**
- Direct actuator control without restarting simulation
- Predictive comfort optimization
- Autonomous Physical AI system
- Dynamic decision making instead of static scheduling

TECHNICAL APPROACH

TECHNOLOGIES USED

Programming

- Python 3.12
- Object-Oriented Programming (OOP)

AI & Machine Learning

- Llama 3.1 8B
- Groq API
- OpenAI Tool Calling
- Rule-Based Predictive Guardrails

Building Simulation

- EnergyPlus 25.2
- pyenergyplus Python API
- DOE Reference Small Office Building
- Chicago TMY3 Weather Dataset (EPW)

Dashboard & Visualization

- Streamlit
- Matplotlib
- Interactive KPI Dashboard

Data Management

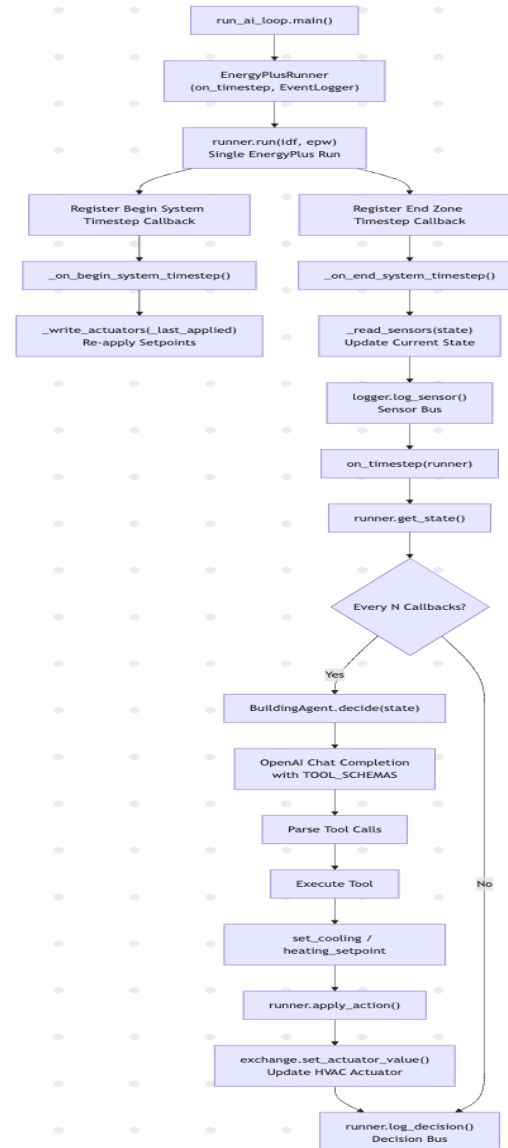
- CSV Logging
- JSONL Event Logs
- Real-Time Sensor Logging

Configuration & Environment

- python-dotenv
- Environment Variables (.env)

Methodology and process for implementation

System architecture: The architecture integrates EnergyPlus, the Python API, an LLM-based Building Agent, OpenAI Tool Calling, and live HVAC actuators to enable autonomous closed-loop building energy optimization.



IMPLEMENTATION METHODOLOGY

Flow Summary

- EnergyPlus starts the building simulation.
- Live sensor data is collected from building zones.
- The Building Agent (LLM) analyzes the current building state.
- The LLM invokes HVAC control tools through OpenAI Tool Calling.
- Predictive guardrails validate every AI decision.
- Approved cooling/heating setpoints are applied to live EnergyPlus actuators.
- Updated building conditions are monitored continuously.
- All sensor readings and AI decisions are logged and displayed on the dashboard.

FEASIBILITY AND VIABILITY

FEASIBILITY AND VIABILITY

Feasibility

- Built on **EnergyPlus 25.2**, an industry-standard building energy simulator.
- Uses the **EnergyPlus Python API** for real-time HVAC actuator control.
- Compatible with existing HVAC and Building Management Systems (BMS).
- No physical hardware modifications required.
- Scalable from single-zone to multi-zone commercial buildings.

Challenges

- Maintaining occupant comfort while reducing energy usage.
- LLM latency and API rate-limit constraints.
- Preventing frequent HVAC setpoint oscillations.
- Balancing energy savings with thermal comfort.

Solutions

- Intelligent callback scheduling to reduce unnecessary AI requests.
- Predictive guardrails with comfort-band enforcement.
- Step-size limiting for smooth HVAC adjustments.
- Automatic heuristic fallback when AI is unavailable.
- Continuous logging and monitoring for reliable operation.

Metric

Result

Baseline Energy	1759.3 kWh
AI-Controlled Energy	1696.8 kWh
Energy Savings	3.6%
Occupied Comfort	66.8%
Control Method	Live LLM + EnergyPlus Actuators

Key Advantages

- Real-time closed-loop AI control
- Dynamic multi-zone HVAC optimization
- Industry-standard simulation platform
- Fully autonomous decision-making
- Energy-efficient and scalable architecture

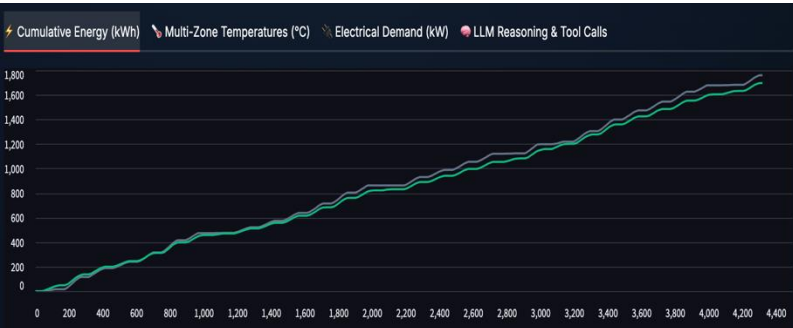
ARTIFACTS

Project Components

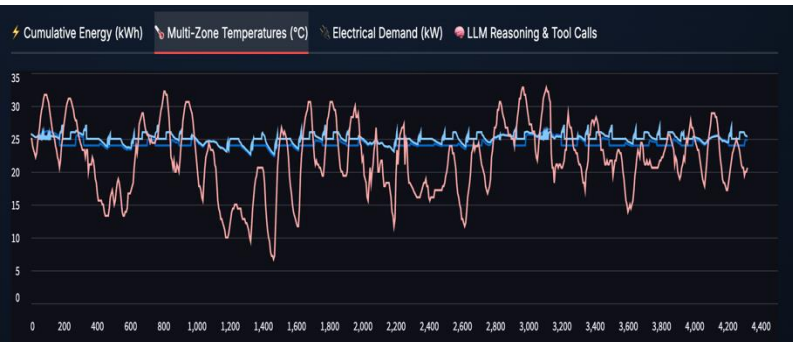
Source Code

- EnergyPlus Runner
- Building Agent
- Tool Calling Framework
- Dashboard
- Logger
- Carbon Module

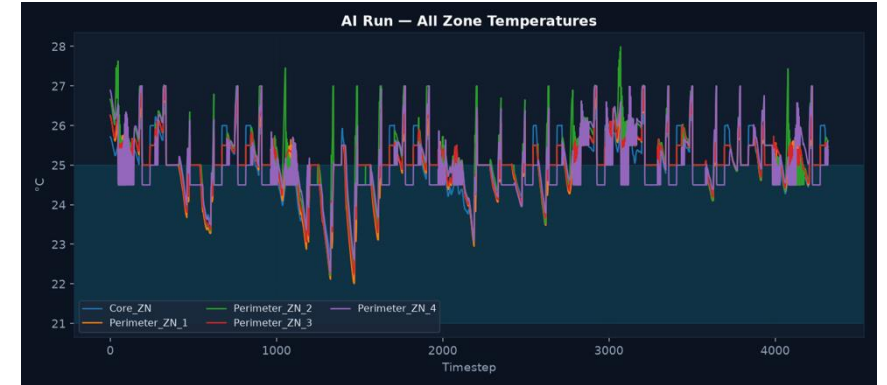
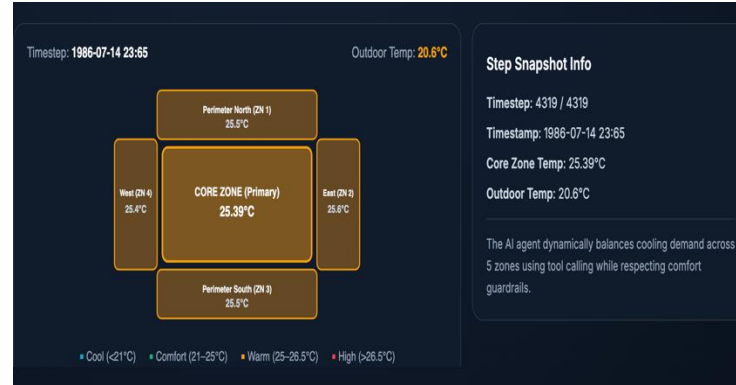
Energy Consumption Graph



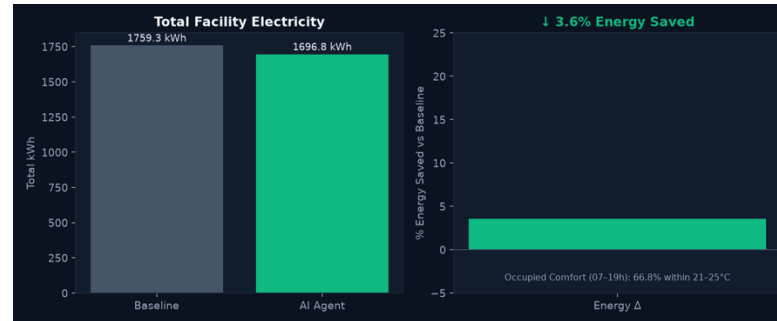
Temperature Graph



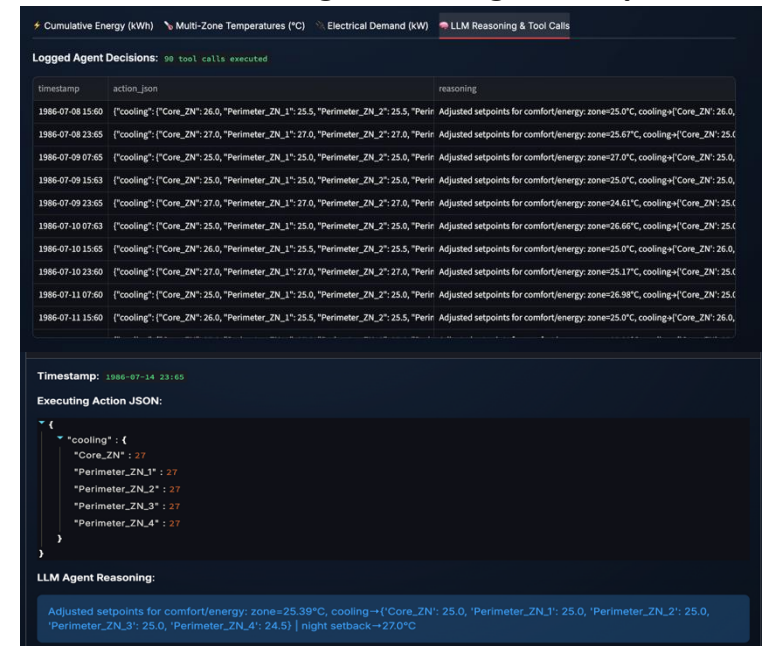
Multi-zone Dashboard



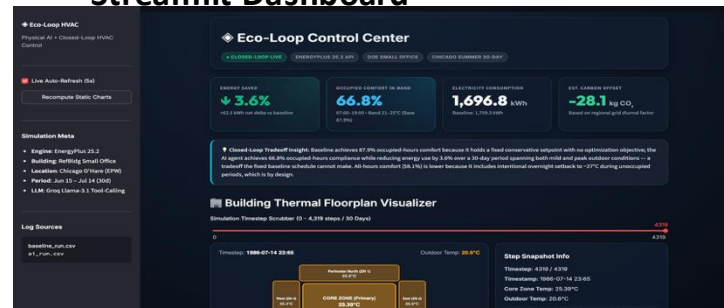
Savings Summary



Live Decision Logs and AI Agent Output



Streamlit Dashboard



RESEARCH AND REFERENCES

Research Areas

- Building Energy Simulation and HVAC Optimization
- Physical AI & Closed-Loop Control Systems
- Large Language Models (LLMs) for Autonomous Decision Making
- Smart Building Energy Management Systems (BMS)
- Sustainable and Energy-Efficient Building Operations

Technical References

- [EnergyPlus 25.2 Documentation](#)
- [EnergyPlus Python API \(pyenergyplus\)](#)
- [OpenAI Tool Calling Documentation](#)
- [Groq Developer Documentation](#)
- [Streamlit Documentation](#)

Standards & Datasets

- DOE Reference Building – Small Office Model
- Chicago TMY3 Weather Dataset (EPW)
- ASHRAE Thermal Comfort Guidelines (21–25° C Comfort Band)

Project Repository

GitHub: <https://github.com/cavxn/Eco-Loop-HVAC>

Key References:

- U.S. Department of Energy (DOE),
- EnergyPlus Documentation, ASHRAE Standards,
- OpenAI API Documentation,
- Groq Developer Platform.

All simulations were conducted using DOE Reference Building models with Chicago TMY3 weather data in EnergyPlus 25.2.